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RESEARCH ARTICLE

A Knowledge-Based Probabilistic Framework for Enhancing Reliability in Healthcare Appointment Scheduling

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ABSTRACT Missed appointments in outpatient healthcare settings continue to be a widespread operational and clinical challenge, resulting in inefficiencies, disrupting care continuity, and contributing to financial losses. Addressing this problem requires analytical frameworks that extend beyond descriptive reporting toward interpretable, decision-oriented models. This study develops a knowledge-based probabilistic framework that integrates advanced machine learning (ML), probabilistic graphical models, and causal inference to predict and mitigate nonattendance. Leveraging 110 526 outpatient records from Brazilian healthcare facilities, the framework applies a structured pipeline that combines systematic feature selection, hybrid class balancing, and model training within a stratified nested cross-validation design to ensure unbiased performance estimation and reduce overfitting. The best-performing model achieved an accuracy of 0.93, an AUC of 0.83, and a macro-F1 of 0.83, surpassing benchmark approaches and providing a foundation for subsequent analyses. Building on these predictive results, Bayesian belief networks (BBNs) were employed to capture probabilistic dependencies among patient, temporal, and appointment-related variables, thereby enhancing interpretability, supporting scenario-based reasoning, and providing the foundation for causal analyses. In the final stage, average treatment effect (ATE) estimation and uplift modeling were applied to evaluate the impact of SMS reminders. Results demonstrate an average 15% reduction in nonattendance probability, with the strongest uplift observed among patients with repeated missed visits. This reduction corresponds to an estimated annual saving of approximately \$670 000 in unused clinical capacity. Collectively, the integration of predictive modeling, probabilistic knowledge representation, and causal inference provides a coherent framework that is both analytically sound and operationally relevant, offering actionable tools to optimize scheduling, allocate resources efficiently, and enhance patient access.

INDEX TERMS Probabilistic risk assessment, Bayesian belief networks, machine learning for operations, causal inference, healthcare operations management, knowledge-based analysis.

I. INTRODUCTION

Patient no-shows for scheduled medical appointments represent a significant and persistent challenge for healthcare

systems worldwide, with major implications for service quality, resource utilization, and patient outcomes. The phenomenon of appointment non-attendance, wherein patients fail to appear for scheduled consultations without prior notification, has been extensively documented across diverse healthcare settings, with reported rates varying dramatically

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from 3% to as high as 80%, depending on the clinical context, patient demographics, and geographical location [1]. This universal issue transcends national boundaries and healthcare system structures, posing a global challenge that necessitates innovative and practical solutions to mitigate its multifaceted negative consequences on healthcare providers and their patients [2].

The financial burden of missed appointments is concerning in an era characterized by escalating healthcare costs and constrained resources. Recent economic analyses have estimated that patient no-shows cost the healthcare industry approximately \$150 billion annually in the United States alone [3], with individual healthcare providers experiencing revenue losses averaging \$200 per missed appointment [4]. These financial losses stem primarily from the underutilization of expensive medical equipment, inefficient allocation of clinical staff time, and reduced operational capacity, all of which contribute to diminished productivity and increased per-patient costs [5]. In specialized departments such as radiology, where complex imaging equipment represents substantial capital investments, the financial impact of appointment non-attendance is further amplified, with a single radiology department potentially experiencing annual revenue losses exceeding \$89,000 due to a no-show rate of 12% [6].

Recent methodological developments in predictive analytics underscore the need for evaluation to extend beyond discrimination to assess calibration and clinical utility. Poor calibration can mislead clinical decisions; therefore, reliability curves and Brier scores should accompany performance reports [7]. Decision curve analysis quantifies net benefit across threshold probabilities and clarifies whether prediction models justify interventions such as overbooking or targeted reminders [7]. Evidence from a rapid review of predictive model-based reminders suggests that text messages and phone calls can reduce no-show rates by approximately 9% and 39%, respectively [8]. These insights motivate the integration of calibration, net benefit assessment, and causal analysis into a cohesive framework for addressing non-attendance.

Beyond these economic considerations and methodological insights, patient no-shows trigger a chain of operational inefficiencies that compromise the quality and accessibility of healthcare services. The unexpected gaps in clinical schedules created by non-attendance disrupt workflow continuity, complicate resource planning, and extend waiting times for all patients seeking care [9]. This event creates contradictory situations wherein healthcare systems experience underutilization of available resources and extended waiting periods for patients requiring services [10]. The resulting scheduling inefficiencies reduce the overall capacity of healthcare systems to meet patient needs and contribute to decreased satisfaction among patients and healthcare providers, thereby undermining the quality of the therapeutic relationship that is fundamental to effective care delivery [11].

The consequences of appointment non-attendance extend beyond operational and financial considerations to directly impact patient health outcomes. When patients fail to attend scheduled appointments, they experience discontinuity in their care processes, delayed diagnoses, interrupted treatment regimens, and compromised disease management, particularly for chronic conditions requiring regular monitoring and intervention. This disruption in the continuity of care has been associated with poorer control of chronic diseases, delayed presentation to care, and increased utilization of emergency services for conditions that could have been addressed more effectively and efficiently in primary care settings [11]. Furthermore, the relationship between appointment non-attendance and health outcomes exhibits a concerning socioeconomic gradient, with vulnerable populations—including those of lower socioeconomic status, those with public insurance, and those facing transportation barriers—disproportionately affected by both higher rates of no-shows and the resulting adverse health consequences [12].

The anatomy of appointment non-attendance is multifactorial and complex, including a diverse array of patient-related, provider-related, and environmental factors that interact dynamically to influence attendance behavior. Empirical investigations have identified various determinants of no-show behavior, including forgetfulness, scheduling conflicts, transportation difficulties, financial constraints, insurance coverage issues, language barriers, resolution of symptoms, anxiety about medical procedures, and misunderstandings regarding the importance of appointments [6], [11]. Additionally, systemic factors such as long waiting times between scheduling and the actual appointment, and employing a one-size-fits-all approach that overlooks the differences between providers and patients, significantly contribute to non-attendance rates. The relative influence of these factors varies considerably across different healthcare settings, patient populations, and types of medical services, necessitating context-specific approaches to understanding and addressing appointment non-attendance [2], [13].

Traditional approaches to mitigating no-shows have primarily relied on reminder systems, including telephone calls, text messages, and emails, which have demonstrated varying degrees of effectiveness in reducing non-attendance rates. For instance, Drabkin et al. reported a substantial reduction in no-show rates, from 20.99% to 7.07% following the implementation of telephone reminders [14]. However, these conventional interventions often employ a one-size-fits-all approach that overlooks the heterogeneity in no-show risk across different patient segments and appointment types. Moreover, such universal reminder systems may represent an inefficient allocation of resources, directing interventions toward patients who would have attended their appointments regardless, while potentially missing those at the highest risk of non-attendance who might benefit from more intensive or tailored interventions.

The era of advanced data analytics, machine learning (ML), and artificial intelligence (AI) has created

opportunities for developing more complex, personalized approaches to predicting and preventing appointment non-attendance [15]. Recent years have witnessed a hype of research exploring the application of predictive modeling techniques to identify patients at elevated risk of no-shows, enabling healthcare providers to implement targeted interventions that maximize attendance rates while optimizing resource utilization. These predictive approaches leverage the vast amounts of data captured in electronic health records, appointment scheduling systems, and other digital platforms to identify patterns and relationships that may not be apparent through traditional statistical methods or clinical intuition [11].

ML approaches to no-show prediction encompass a diverse range of methodologies, spanning from traditional statistical techniques such as logistic regression (LR) to more complex algorithms, including random forests (RF), support vector machines (SVM), and neural networks (NN). These methods vary in their computational complexity, interpretability, and predictive performance, with recent comparative analyses suggesting that ensemble methods and deep learning approaches often achieve superior accuracy in identifying patients at risk of non-attendance [11], [15]. However, the relative performance of different predictive models appears context-dependent, influenced by factors such as the specific healthcare setting, the characteristics of the patient population, and the nature and quality of the available data.

A promising development in the field of no-show prediction is the application of interpretable ML approaches that provide accurate forecasts and offer insights into the factors driving those predictions [16]. This approach incorporated mechanisms for handling missing data, a common challenge in healthcare analytics, and provided explanations for individual predictions, thereby enhancing the clinical utility and acceptability of the model. Such interpretable approaches address a critical limitation of “black box” ML, which, despite potentially high predictive accuracy, may face resistance in clinical implementation due to concerns about transparency, accountability, and alignment with clinical reasoning processes [12].

Beyond traditional ML approaches, Bayesian belief networks (BBNs) have surfaced as a valuable methodology for no-show prediction, offering unique advantages in interpretability, uncertainty quantification, and the ability to incorporate domain knowledge. BBNs represent probabilistic relationships between variables through directed acyclic graphs, visually and intuitively representing the factors influencing no-show behavior and their interdependencies. This approach facilitates prediction and supports causal reasoning and scenario analysis, enabling healthcare providers to understand how interventions targeting specific factors might influence attendance outcomes [17]. The explicit representation of uncertainty in BBNs is valuable in healthcare contexts, where decision-making often occurs under incomplete information and inherent variability.

While predictive modeling approaches have demonstrated promise in identifying patients at risk of non-attendance, the ultimate goal of such predictions is to inform interventions that effectively reduce no-show rates and mitigate their negative consequences. In this context, uplift modeling, a specialized form of predictive modeling focused on estimating the incremental impact of interventions on individual outcomes, offers a sophisticated framework for optimizing intervention strategies. By identifying which patients are at high risk of non-attendance and which patients are most likely to respond positively to specific interventions, uplift modeling enables healthcare providers to allocate limited resources more efficiently and effectively [18]. This approach represents a shift from simply predicting outcomes to predicting the causal effect of interventions, thereby supporting more nuanced and personalized approaches to improving appointment attendance.

The integration of predictive analytics into appointment scheduling and patient engagement strategies can transform how healthcare systems address the challenge of no-shows, moving from reactive responses to proactive, data-driven approaches that prevent non-attendance before it occurs. By identifying patients at high risk of no-shows during scheduling, healthcare providers can implement targeted interventions such as personalized reminders, transportation assistance, telemedicine options, or modified scheduling practices to maximize attendance probability. Furthermore, the insights generated through predictive modeling can inform broader system-level changes in appointment scheduling policies, patient communication strategies, and service delivery models, addressing the root causes of non-attendance and enhancing overall system efficiency and effectiveness.

In this paper, a comprehensive investigation of patient no-shows in healthcare settings is presented, employing a multilayered methodology that integrates feature engineering, selection techniques, data balancing, ML prediction models, BBN, and uplift modeling to predict no-show behavior, optimize attendance, and inform data-driven strategies for enhancing operational efficiency and patient engagement. Our approach is built upon and extends existing research in this domain, addressing key limitations and challenges identified in previous studies, while leveraging the complementary strengths of diverse analytical methodologies. Through this integrated approach, healthcare providers aim to be provided with actionable insights and tools for effectively managing the persistent challenge of appointment non-attendance, ultimately improving resource utilization, service quality, and patient outcomes in healthcare delivery systems.

II. RELATED WORKS

The healthcare sector has increasingly turned to advanced predictive analytics to tackle operational challenges, with patient no-shows emerging as an issue that strains resources and disrupts care continuity, prompting researchers to explore various ML techniques to forecast no-show probabilities

to enable proactive interventions and optimize appointment scheduling.

Deina et al. [19] investigated the efficacy of classification algorithms in conjunction with resampling techniques to address the problem of unbalanced datasets in no-show prediction, employing cross-validation to ensure robustness and mitigate bias. Their approach utilized symbolic regression (SR). Instance hardness threshold (IHT) methods surpass traditional ML algorithms, such as K-Nearest Neighbors (KNN) and SVM, by achieving sensitivity metrics exceeding 0.94. However, the absence of a comprehensive comparison with tree-based and ensemble methods in their study highlights the need for further research to fully understand the comparative advantages of these techniques in this context [19].

Complementing this work, Joseph et al. conducted a comparative analysis of multiple prediction models, including LR, Decision Tree (DT) Classifier, RF, SVM, and Gradient Boosting (GB), with the latter demonstrating superior performance as evidenced by an F1 score of 0.84, yet despite the span of models evaluated, the study notably lacked the implementation of cross-validation and balancing techniques, suggesting potential avenues for methodological refinement in future investigations [20].

The challenge of dataset imbalance, a common obstacle in no-show prediction due to the disproportionate representation of show versus no-show instances, has been a focal point for researchers seeking to enhance model accuracy and generalizability, with Krishnan and Sangar [21] tackling this issue by evaluating an array of sampling techniques, ranging from Random Over Sampling and Synthetic Minority Oversampling Technique to Edited Nearest Neighbor and Condensed Nearest Neighbor, to rectify biases favoring the majority class, their findings derived from a Decision Tree Classifier identifying Edited Nearest Neighbor as the most effective method for improving recall. However, the study's reliance on a single ML model and a limited dataset highlights the need for broader methodological exploration [21].

In pursuit of optimizing predictive performance, researchers have also explored advanced methods for variable selection, with genetic algorithms and simulated annealing emerging as promising approaches, as exemplified by Simsek et al., who introduced a four-step methodology that integrates genetic algorithms (GA) and simulated annealing (SA) for feature selection, coupled with techniques to address data imbalance such as Synthetic Minority Oversampling Technique (SMOTE) and Random Under Sampling (RUS), their application of various ML models, including RF and ANN, resulting in the ANN-GA-RUS model achieving an ROC of 0.844, thereby demonstrating the potential of hybrid approaches in stratifying patient risk [22].

Extending the methodological range, BBNs have been leveraged to model the probabilistic relationships underlying no-show behavior, offering a framework for both prediction and inference, as demonstrated in a subsequent study by Simsek et al. [23], which presented a five-step probabilistic methodology incorporating Extreme Gradient Boosting

(XGboost), Particle Swarm Optimization (PSO), and GA for variable selection, alongside SMOTE and RUS for data balancing, their comparison of Tree-Augmented Naive Bayes (TAN) with other models revealing the TAN-GA & PSO-RUS configuration yielding an ROC of 0.828 and a sensitivity of 0.785. However, concerns regarding the preservation of original data distribution in k-fold cross-validation warrant cautious interpretation of these metrics [23].

Beyond refining predictive accuracy, recent research has begun to explore how predictive insights can be translated into actionable interventions, with uplift modeling emerging as a promising framework for personalized no-show mitigation strategies, its pioneering applications in healthcare, such as those by Smith et al. [48] in diabetes management and Brown et al. [49] in mental health interventions, having demonstrated substantial improvements in treatment effectiveness through personalized strategies, the two-model approach of uplift modeling, which estimates the net gain from interventions by comparing treated and untreated patient outcomes, holding particular promise for addressing no-shows by identifying patients most likely to benefit from specific reminders or support mechanisms, yet despite its successes in other healthcare domains, the application of uplift modeling to hospital appointment attendance remains underexplored, presenting a fertile ground for research aimed at reducing no-show rates through precision interventions [24], [25], [26], [27].

III. METHODOLOGY

In this study, a comprehensive knowledge-based framework is proposed that integrates three knowledge representation paradigms. The approach combines feature selection methods, data balancing techniques, and ML algorithms to construct a robust BBN that captures the probabilistic interactions among diverse input variables and their influence on the no-show outcome. This enhanced methodology, presented in Figure 1 and Algorithm 1, serves as a foundation for intervention testing, leveraging uplift modeling.

A. ALGORITHMIC FRAMEWORK

The integrated knowledge-based framework operates through a multi-stage pipeline as presented in algorithm 1:

B. DATA COLLECTION AND FEATURE ENGINEERING

The study utilized an open-source Kaggle [28] dataset containing 110,526 records and 14 variables from Brazilian public healthcare providers. Feature engineering was applied to derive additional variables, including temporal factors, patient-specific metrics, and appointment characteristics. Negative lead times were replaced with 0. The final dataset, with its enhanced variables, is presented in Table 1.

C. FEATURE SELECTION

To enhance model generalizability, reduce overfitting, and improve interpretability while achieving computational efficiency, feature selection was performed using three

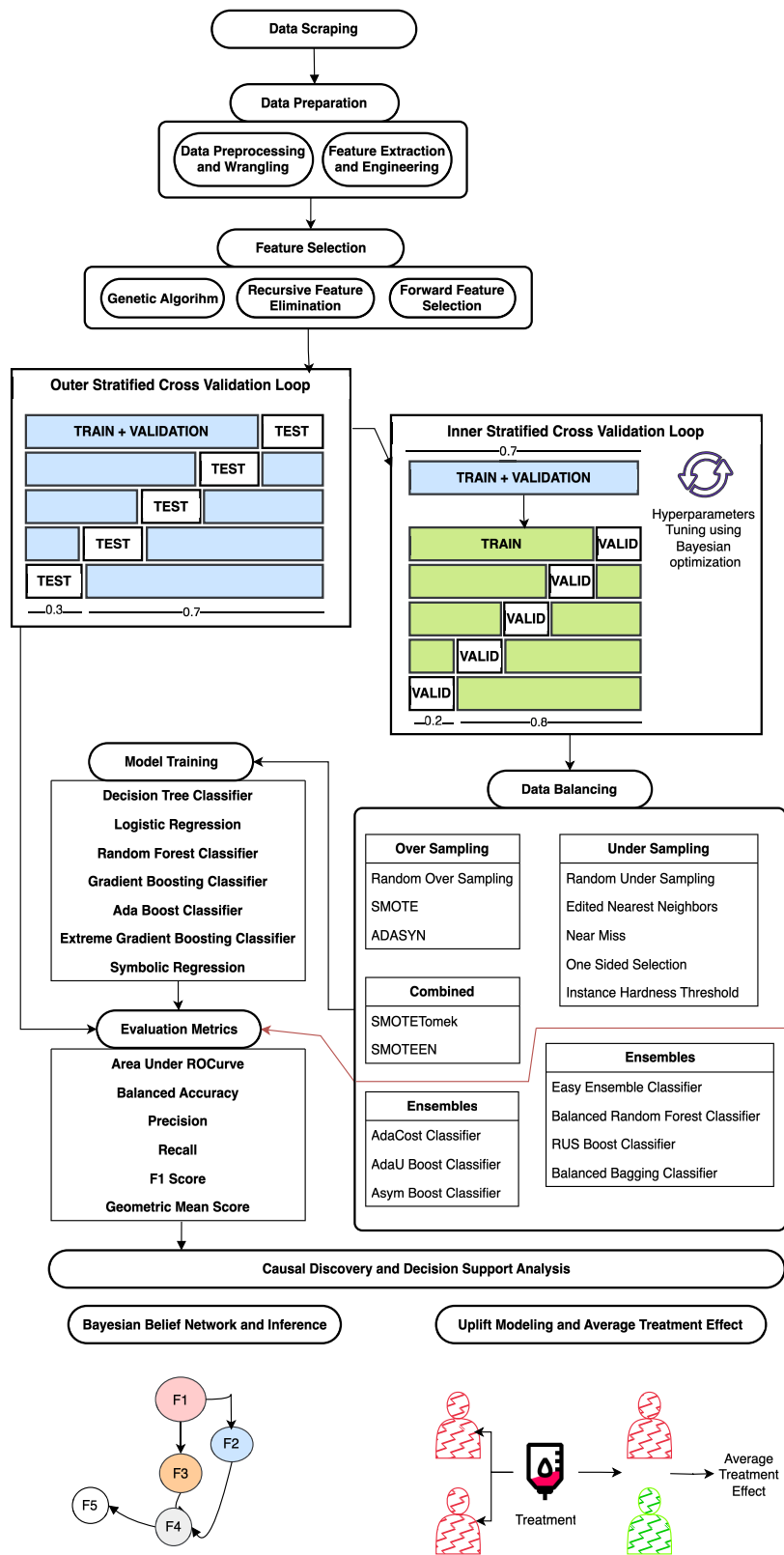


FIGURE 1. Proposed methodology outline.

strategies: Genetic Algorithm (GA) employing population-based search where candidate feature subsets evolved through

mutation, crossover, and tournament-style selection with F1 score fitness evaluation until convergence [29], Recursive

Algorithm 1 Knowledge-Based Prediction Framework

Require: Dataset D , Feature set X , Outcome Y , Treatment indicator T

Ensure: ML model f^* , Probabilistic network G , Effect estimates $\hat{\tau}$

Phase 1: Feature Selection and Preprocessing

$X_{selected} \leftarrow \text{FeatureSelection}(X, Y)$

$D_{balanced} \leftarrow \text{DataBalancing}(D, X_{selected}, Y)$

Phase 2: Machine Learning Knowledge Discovery

Initialize population P_0 of candidate models

for iteration $i = 1$ to I_{max} **do**

 Evaluate performance for each candidate in P_i using inner cross-validation

 Select promising candidates based on validation metrics

 Apply variation operators (crossover, mutation) to generate new candidates

$P_{i+1} \leftarrow$ updated population with new candidates

end for

$f^* \leftarrow$ best performing candidate from final population

 Perform stratified outer cross-validation for unbiased performance estimation

 Conduct hyperparameter tuning using nested cross-validation

 Validate final model using holdout test set

Phase 3: Probabilistic Knowledge Modeling

Discretize continuous variables using clustering methods

$G \leftarrow \text{StructureLearning}(X_{selected}, Y)$

Estimate probability distributions for each network component

Validate network structure using cross-validation

Phase 4: Causal Inference

$\hat{\tau}_{ATE} \leftarrow \text{EstimateAverageEffect}(D, T, Y)$

Build a heterogeneous effect model for subgroup analysis

$\hat{\tau} \leftarrow \text{EstimateUpliftEffects}(D, T, Y, \text{subgroups})$

Validate causal estimates and uplift estimates $\hat{\tau}$ using robustness checks

Return $f^*, G, \hat{\tau}$

Feature Elimination (RFE) utilizing backward-elimination whereby the least important variables were systematically removed based on cross-validated F1 score impact while preserving covariate interactions [30], and Forward Feature Selection (FFS) implementing greedy sequential addition from an empty set by appending variables yielding the most significant incremental F1 score improvement [31]. The comparative evaluation of these methods under identical resampling and validation schemes enabled the optimal selection of feature subsets for subsequent use in both ML classifiers and BBN, ensuring modeling consistency.

1) DATA BALANCE

To evaluate data balancing methods and determine the most effective classifiers, a nested stratified cross-validation scheme was employed. The outer layer used GroupKFold to partition the cohort into five folds, while preserving the proportion of no-show and show appointments. Grouping by patient ensured that all visits from the same individual were confined to a single fold, thereby avoiding information leakage and reducing overfitting. Four folds were designated for training, whereas the fifth served as an independent test set. Within each outer training set, Bayesian optimization guided a stratified five-fold cross-validation search for

TABLE 1. List of variables of the no show dataset.

Variables		Type	Overall Ap- point- ments	Show	No Show	Description
Appoint. ID		NUMC	110526 (100%)	88207 (79.8%)	22319 (20.2%)	Appointment Identification Number
PatientID		NUMC	61744	53730	7669	Patient Identification for grouping
Patient-Related Factors						
Gender	Male	Binary	38687 (35%)	30962 (28%)	7725 (7%)	Patient's Gender (Male, Female)
	Female		71839 (65%)	57245 (51.8%)	14594 (13.2%)	
Age		NUMC	37.1 ± 23.1	37.8 ± 23.3	34.3 ± 22	Age of patients
Neighbourhood		82 Categs	-	-	-	Geographic location or neighborhood where the patient resides
Insurance	No	2 Categs	99665 (90.2%)	79924 (72.3%)	19741 (17.9%)	Presence or absence of insurance coverage
	Yes		10861 (9.8%)	8283 (7.5%)	2578 (2.3%)	
Hyper- tension	No	2 Categs	88725 (80.3%)	70178 (63.5%)	88725 (16.8%)	Whether or not a patient is diagnosed with high blood pressure
	Yes		21801 (19.7%)	18029 (16.3%)	21801 (3.4%)	
Diabetes	No	2 Categs	102583 (92.8%)	81694 (73.9%)	20889 (18.9%)	Whether or not a patient is diagnosed with blood sugar
	Yes		7943 (7.2%)	6513 (5.9%)	1430 (1.3%)	
Alcohol- ism	No	2 Categs	107166 (97%)	85524 (77.4%)	21642 (19.6%)	whether a patient has a history or current diagnosis of alcoholism
	Yes		3360 (3%)	2683 (2.4%)	677 (0.6%)	
Handi- cap	No	5 Categs	108285 (98%)	86373 (78.2%)	21912 (19.8%)	Presence or absence of handicaps or disabilities
	1		2042 (1.75%)	1676 (1.5%)	366 (0.25%)	
	2		183 (0.165%)	146 (0.13%)	37 (0.035%)	
	3		13 (0.12%)	10 (0.009%)	3 (0.003%)	
	4		3 (0.003%)	2 (0.002%)	1 (0.001%)	
Appointment Related Factors						
SMS Received	No	2 Categs	75044 (67.9%)	62509 (56.6%)	12535 (11.3%)	Whether an SMS reminder was sent before the appointment day
	Yes		35482 (32.1%)	25698 (23.3%)	9784 (8.8%)	
Reg. Year	2015	2 Categs	62 (0.06%)	43 (0.04%)	19 (0.02%)	The Calendar year in which the appointment registration/booking occurred.
	2016		110464 (99.94%)	88164 (79.76%)	22300 (20.18%)	
Reg. Season	Autumn	4 Categs	62 (0.06%)	43 (0.04%)	19 (0.02%)	The period of year/season in which the appointment registration/booking occurred.
	Spring		108175 (97.87%)	86593 (78.35%)	86593 (19.53%)	
	Winter		2289 (2.04%)	1571 (1.4%)	1571 (0.65%)	
Reg. Month	1	12 Categs	60 (0.06%)	42 (0.04%)	18 (0.02%)	The Calendar month in which the appointment registration/booking occurred.
	2		281 (0.25%)	199 (0.18%)	82 (0.07%)	
	3		3614 (3.3%)	2418 (2.2%)	1196 (1.1%)	
	4		25339 (22.9%)	17490 (15.8%)	7849 (7.1%)	
	5		67421 (61%)	55652 (50.4%)	11769 (10.6%)	
	6		13749 (12.4%)	12363 (11.2%)	1386 (1.2%)	
	12		61 (0.06%)	42 (0.04%)	19 (0.02%)	
Reg. Week		NUMC	19 ± 3	19 ± 3	17 ± 3	The Calendar week number in which the appointment registration/booking occurred.
Reg. Day		NUMC	14 ± 9	14 ± 9	15 ± 9	The Calendar day of the month for the appointment registration/booking.

TABLE 1. (Continued.) List of variables of the no show dataset.

Reg. Day of Week	Friday	7 Categs	18915 (17%)	15028 (13.5%)	3887 (3.5%)	The Calendar day of the week in which the appointment registration/booking occurred.
	Monday		23084 (20.9%)	18523 (16.8%)	4561 (4.1%)	
	Satur-day		24 (0.022%)	23 (0.021%)	1 (0.001%)	
	Thurs-day		18073 (16.4%)	14373 (13.0%)	3700 (3.3%)	
	Tuesday		26168 (23.7%)	20877 (18.9%)	5291 (4.8%)	
	Wednes-day		24262 (22.0%)	19383 (17.5%)	4879 (4.4%)	
Reg. Hour		NUMC	10 ± 3	10 ± 3	11 ± 3	The Calendar hour of the day in which the appointment registration/booking occurred.
Reg. Time	Day Time	2 Categs	110423 (99.9%)	88135 (79.7%)	22288 (20.2%)	The day period in which the appointment registration/booking occurred.
	Night Time		103 (1%)	72 (0.07%)	31 (0.03%)	
Appoint. Year	2016	2 Categs	110526 (100%)	88207 (79.8%)	22319 (20.2%)	The Calendar year in which the appointment was registered or booked.
Appoint. Season	Spring	4 Categs	110526 (100%)	88207 (79.8%)	22319 (20.2%)	The period of year/season in which the appointment was registered or booked.
Reg. Month	4	12 Categs	3235 (3%)	2602 (2.4%)	633 (0.6%)	The Calendar month in which the appointment was registered or booked.
	5		80841 (73.1%)	64037 (57.9%)	16804 (15.2%)	
	6		26450 (23.9%)	21568 (19.5%)	4882 (4.4%)	
Appoint. Week		NUMC	21 ± 2	21 ± 2	20 ± 2	The Calendar week number in which the appointment was registered or booked.
Appoint. Day		NUMC	12 ± 9	13 ± 9	12 ± 8	The Calendar day of the month the appointment was registered or booked.
Reg. Day of Week	Friday	7 Categs	19019 (17.2%)	14982 (13.6%)	4037 (3.7%)	The Calendar day of the week in which the appointment was registered or booked.
	Monday		22714 (20.6%)	18024 (16.3%)	4690 (4.2%)	
	Satur-day		39 (0.035%)	30 (0.027%)	9 (0.008%)	
	Thurs-day		17247 (15.6%)	13909 (12.6%)	3338 (3.0%)	
	Tuesday		25640 (23.2%)	20488 (18.5%)	5152 (4.7%)	
	Wednes-day		25867 (23.4%)	20774 (18.8%)	5093 (4.6%)	
Appoint. Hour		NUMC	-	-	-	The Calendar hour of the day the appointment was registered or booked.
Appoint. Time		NUMC	-	-	-	The day period the appointment was registered or booked.
Lead Time (days)		NUMC	9.5 ± 15	8.2 ± 14.3	15 ± 16.5	The duration between the booking date and the actual appointment day
Num Visits		NUMC	1.28 ± 4	1.33 ± 4.22	1.32 ± 2.4	The total count of appointments a patient has had before the current one.
Prev Show		NUMC	1 ± 3.8	-	-	The count of past appointments where the patient successfully attended their scheduled appointments.
Prev NoShow		NUMC	0.23 ± 1.16	-	-	The count of past appointments where the patient did not attend their scheduled appointments.
Time Between Appoint.		NUMC	6.5 ± 10.3	6.4 ± 10.3	6.8 ± 10.3	The duration in days between two consecutive appointments scheduled for the same patient.
Time Between Reg		NUMC	6.7 ± 15.1	7 ± 15.1	5.1 ± 14.6	The duration between two consecutive registrations for appointments made by the same patient.

hyperparameters across a suite of algorithms, including decision trees, random forests, gradient boosting, adaptive boosting, extreme gradient boosting, and symbolic regression classifiers [32], [33]. The optimization objective combined macro F1 with area under the receiver operating characteristic curve, and stratification in the inner loop maintained class proportions during hyperparameter tuning.

Model assessment extended beyond discrimination metrics. Across outer folds, precision, recall, macro- and micro-averaged F1, macro-, AUC, and G-means were calculated. Calibration was evaluated using reliability diagrams and the Brier score, and clinical utility was quantified through decision curve analysis, which estimates net benefit across clinically reasonable threshold probabilities [46], [47]. This comprehensive evaluation yields a nuanced picture of performance for imbalanced outcomes and provides more informative insight than accuracy alone.

D. BAYESIAN BELIEF NETWORK

In the context of hospital no-show predictions, BBNs model complex dependencies among variables. The methodology involves constructing a graphical model representing probabilistic relationships, with nodes symbolizing influential features and edges signifying probabilistic dependencies.

BBNs are based on Bayes' theorem:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)} \quad (1)$$

The joint probability distribution of the network is expressed as:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i | \text{Parents}(X_i)) \quad (2)$$

Multiple structure learning approaches were employed to construct BBNs, including the Chow-Liu algorithm, Max-Min Hill Climbing (MMHC), Tree Augmented Naive Bayes (TAN), Greedy Hill Climbing (GHC), Tabu search, and Naive Bayes classifiers [34], [35], [36], [37], [38], [39]. Continuous variables were discretized using k-means clustering, MDLP, and CAIM [40], [41], [42]. BBNs are advantageous for no-show prediction because they can capture complex interdependencies among variables and exhibit resilience to missing data, a common challenge in clinical datasets [43].

To prevent information leakage, discretizations and network structure learning were performed exclusively on the training segments of each cross-validation fold. The resulting bins and structures were then applied unchanged to the validation and test data. Comparisons of edge sets and posterior distributions across k-means, MDLP, and CAIM indicated that although marginal probabilities varied slightly, the presence and direction of edges remained stable. This suggests that the causal inferences drawn from the BBNs are robust to the choice of the discretization scheme and affirms the utility of the selected structure learning algorithms.

E. UPLIFT MODELING AND CAUSAL EFFECT ESTIMATION

Uplift modeling was employed to quantify the individual-level effect of the SMS reminder on attendance at appointments. First, a regularized LR model estimated the propensity score $e(X) = \Pr(T = 1 | X)$ using all covariates available in the dataset—demographic factors (age, gender, neighborhood, insurance status), comorbidities (hypertension, diabetes, alcoholism), temporal attributes (registration season, month, week, day, hour and period, appointment season, month and week), and historical utilization variables (prior shows, prior no-shows, number of visits, lead time, time between appointments and registrations). Hyperparameters were tuned via cross-validation, and one-to-one nearest neighbor matching on the logit of the propensity score, with a caliper restriction (without replacement), was used to construct balanced treatment and control groups. Covariate balance was verified by computing standardized mean differences for each covariate before and after matching and by inspecting propensity score histograms; values below 0.10 indicated strong balance, and values up to 0.25 were considered acceptable.

After confirming adequate overlap, two conditional outcome models were used to predict the probability of a no-show for each individual under both treatment and control conditions. The individual uplift score U was then calculated as the difference between these probabilities:

$$U = P(\text{No Show} | X, T = 1) - P(\text{No Show} | X, T = 0),$$

where a higher positive U indicates that the reminder is expected to reduce the likelihood of a no-show for that individual, model performance was evaluated using metrics such as the area under the uplift curve (AUUC) and the uplift Gini coefficient, which integrate the uplift curve to reward models that rank individuals correctly by potential benefit [44], [45]. Patients with the highest positive uplift scores were prioritized for intervention, and the allocation strategy was iteratively refined to maximize attendance while maintaining balance across covariates.

IV. RESULTS

This study investigated no-show prediction in healthcare settings using a comprehensive approach integrating feature selection, data balancing, ML algorithms, BBN, and uplift modeling. The goal was to develop a robust predictive framework for identifying patients at risk and providing actionable insights for targeted interventions.

The correlation between features and the target variable (hospital no-show) was studied using correlation analysis techniques. Figure 2 represents the correlation funnel plot, showing features arranged in descending order of their absolute correlation values with the target variable. This visualization technique was adapted to aid in feature selection, helping to reduce dataset dimensionality and improve model interpretability while accounting for multiple testing corrections.

TABLE 2. Comparison of the selected features in GA, RFE, and SFS.

Variables	GA	RFE	FFS
Age		*	*
Gender	*		
Neighbourhood	*	*	*
Insurance	*	*	*
Hypertension			
Diabetes	*		*
Alcoholism	*	*	*
SMS_received		*	*
RegistrationSeason	*	*	
RegistrationMonth		*	*
RegistrationWeek		*	*
RegistrationDay	*		
RegistrationDate		*	
RegistrationHour		*	
AppointmentMonth			*
AppointmentWeek		*	*
AppointmentDate		*	
LeadTime	*	*	*
NumVisits	*	*	
PrevShow		*	*
PrevNoShow	*	*	*
TimeBetweenApp	*	*	*
TimeBetweenReg		*	*

Additional feature selection techniques were employed after identifying relevant features through correlation analysis and the correlation funnel plot. Table 2 presents features selected by GA, RFE, and FFS, with asterisks (*) indicating chosen features by each method. This multi-method approach provided diverse perspectives on the relevance of features. Selected features were subsequently utilized in modeling and analysis steps to enhance the performance and interpretability of the no-show prediction model.

After feature selection, the dataset's imbalanced nature was observed, with 79.8% of appointments being attended. Data balancing techniques were employed to address this issue. Various ML models, including DT, RF, GBoost, AdaBoost, XGBoost, and SR classifiers, were evaluated for no-show prediction using standard performance metrics such as accuracy, precision, recall, macro F1-score, G-means, and AUROC. Table 3 summarizes the best model performance, considering the impact of different feature selection and balancing methods on predictive outcomes, with each row representing the optimal combination of techniques.

The grouped cross-validation protocol yielded macro and micro AUPRC values of 0.69 and 0.72, respectively, for the SFS + SMOTEEN + SR model, indicating a strong ability to identify rare no-show instances. Calibration analysis produced a Brier score of 0.11, with only slight overconfidence in the highest risk decile. As shown in Table 3, this model also achieved an AUROC of 0.83, G-Means of 0.83, precision of 0.87, recall of 0.80, macro

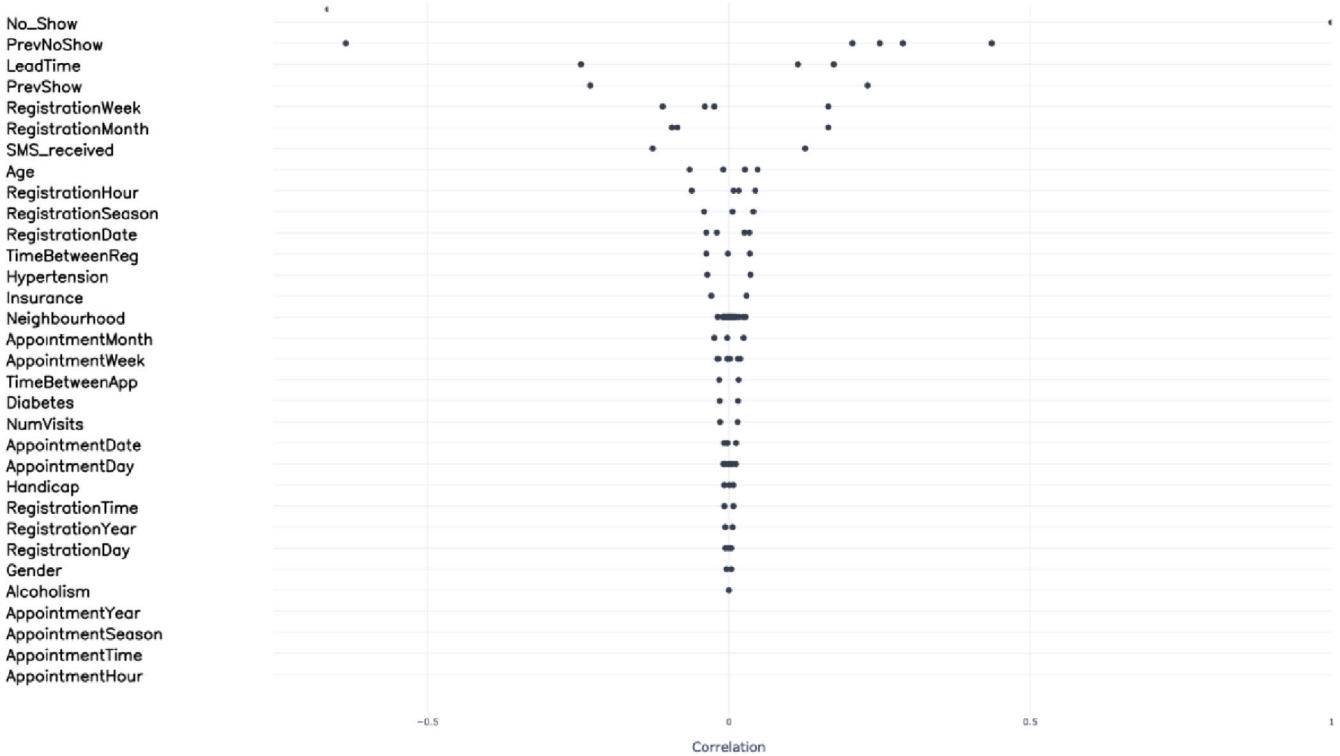


FIGURE 2. Correlation funnel plot.

TABLE 3. Best models performance using different feature selection and balancing methods.

Selection Method	Balancing Method	ML Model	Metrics					
			ACC	AUC	G Means	Precision	Recall	F1 Score
Original Data	ROS	XGBoost	0.92	0.82	0.82	0.86	0.79	0.82
GA	SMOTEEN	GB	0.96	0.81	0.8	0.84	0.76	0.8
RFE	SMOTE	XGBoost	0.91	0.81	0.79	0.82	0.76	0.79
SFS	SMOTEEN	SR	0.93	0.83	0.83	0.87	0.80	0.83

F1 score of 0.83, and accuracy of 0.93. These metrics demonstrate that SFS + SMOTEEN + SR offers the best overall balance across evaluation criteria, outperforming alternative pipelines, while the critical-difference diagram confirms it ranks first among comparable methods. Together, the combined discrimination, calibration, and net-benefit results indicate that this model is both accurate and clinically useful.

Our approach demonstrates competitive performance in the benchmark comparison shown in Table 4, compared to the models that were discussed in the Related Work section. The critical difference diagram generated using the post hoc Conover-Friedman test visually represents the average performance ranking. It shows that our model’s average score ranks first in comparison with other models that use a similar methodology. Our model achieves the highest accuracy (0.93) and demonstrates robust precision (0.87) and F1 score (0.83). Our slightly lower recall (0.8) is counterbalanced by superior

precision, indicating a more practical trade-off between false positives and negatives. This balance, combined with our use of balancing techniques, suggests our model offers enhanced generalizability and real-world applicability compared to others.

To ensure a fair comparison, all baseline models reproduced from prior studies were reimplemented within our pipeline, using the same feature sets, data balancing techniques, and cross-validation splits. Each model was evaluated across ten random seeds to account for stochastic variation. The Friedman test was applied to compare the median ranks of models across metrics; the resulting $\chi^2_5 = 9.84$ with $p = 0.08$ indicated marginal evidence of differences at the 0.05 level. Post hoc Conover tests with Holm-Bonferroni correction were conducted, and effect sizes were reported as Cliff’s delta. The updated ranking plot in Figure 3 displays mean ranks with 95% confidence intervals across seeds.

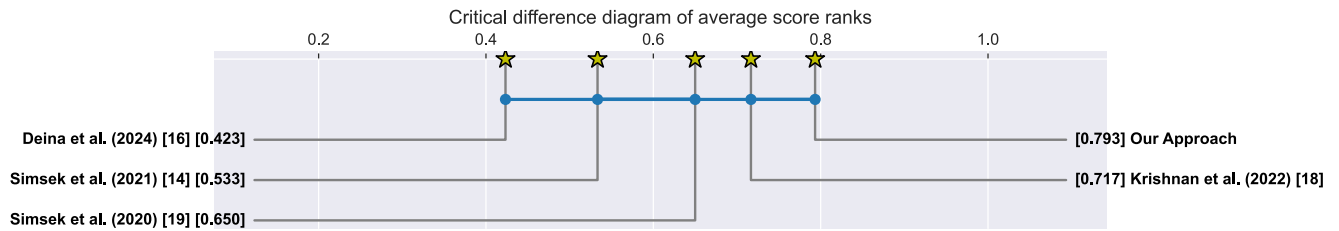


FIGURE 3. Posthoc conover-friedman average metrics ranking.

TABLE 4. Benchmark between our approach and others using the same dataset.

	ACC	AUC	Precision	Recall	F1
Deina et al. (2024) [19]	0.67	0.77	0.36	0.94	0.52
Krishnan et al. (2022) [21]	-	0.51	0.78	0.998	0.89
Simsek et al. (2020) [22]	0.74	0.84	-	0.78	-
Simsek et al. (2021) [23]	0.73	0.83	-	0.79	-
Our Approach	0.93	0.83	0.87	0.8	0.83

A BBN was constructed using significant features from the best-performing combination to model complex probabilistic relationships influencing no-shows. The Chow-Liu structure learning method, combined with k-means discretization, as presented in Figure 4, was found to provide the best performance. This combination achieved 97.4% accuracy in predicting showcases and 82% accuracy for no-show cases. The Chow-Liu algorithm effectively captured intricate relationships among significant features, while k-means discretization facilitated the modeling of non-linear dependencies in continuous variables.

The BBN structure, illustrated in Figure 4, comprises interconnected nodes and edges representing relationships between variables influencing appointment attendance. The central red node, labeled “No-Show,” represents the target variable. It is directly influenced by “PrevNoShow,” representing the patient’s history of missed appointments, which is influenced by “TimeBetweenApp,” likely representing the interval between appointments. This structure captures the complex relationships among factors that affect no-show behavior.

The “TimeBetweenApp” node influences “TimeBetweenReg,” “PrevShow,” and “AppointmentWeek.” “PrevShow” influences “Age,” which in turn affects “Insurance,” “Diabetes,” and “Alcoholism.” This structure captures the interrelationships between appointment timing, patient history, demographics, and health conditions, providing insights into factors contributing to no-show behavior.

The “AppointmentWeek” node influences “SMS-received,” “AppointmentMonth,” and “RegistrationWeek.” “RegistrationWeek” affects “RegistrationMonth,” which

influences “Neighbourhood.” “Neighbourhood” then impacts “LeadTime.” This structure captures temporal relationships between appointment scheduling, SMS reminders, and geographical factors, providing a comprehensive view of the appointment process and its potential impact on no-show behavior.

The BBN captures the complex relationships among factors that influence appointment no-shows. Sensitivity analysis using BBN inference provides insights into the impact of specific variables on the outcome. Figure 5 demonstrates setting the “No-Show” node to 100% probability, simulating a guaranteed missed appointment. This approach enables the investigation of factors and their likelihood of association with no-show events, providing a deeper understanding of the underlying mechanisms and potential intervention points.

The BBN inference algorithm propagates evidence through the network, updating probabilities based on conditional dependencies. A causation diagram, initiated from the “Neighbourhood” node, was analyzed. Table 5 presents the top 10 neighborhoods with the highest no-show percentages. Factors such as distance from the hospital, insurance status, alcohol consumption history, and chronic disease prevalence were identified as primary drivers of no-shows. This BBN-based analysis enables healthcare organizations to understand complex relationships influencing patient attendance, facilitating the development of effective strategies and targeted interventions to mitigate no-shows and optimize resource allocation.

One of the significant features identified was the receipt of SMS reminders. To evaluate their effectiveness, ATE analysis and uplift modeling were employed. The dataset was divided into control and treatment groups, with Matching Bias criteria applied to address confounding variables. Propensity score matching was utilized to ensure comparability between groups.

The ATE score of -0.149 suggests a 15% average decrease in no-show probability for individuals who receive SMS reminders. An uplift tree model, illustrated in Figure 6, demonstrates that for patients with lead time ≥ 11 days and time between appointments ≥ 17 days, SMS reminders reduce no-show probability by 6.7%. For patients with a previous no-show count of ≥ 3 , the reduction increases to 10%. This analysis provides insights for optimizing

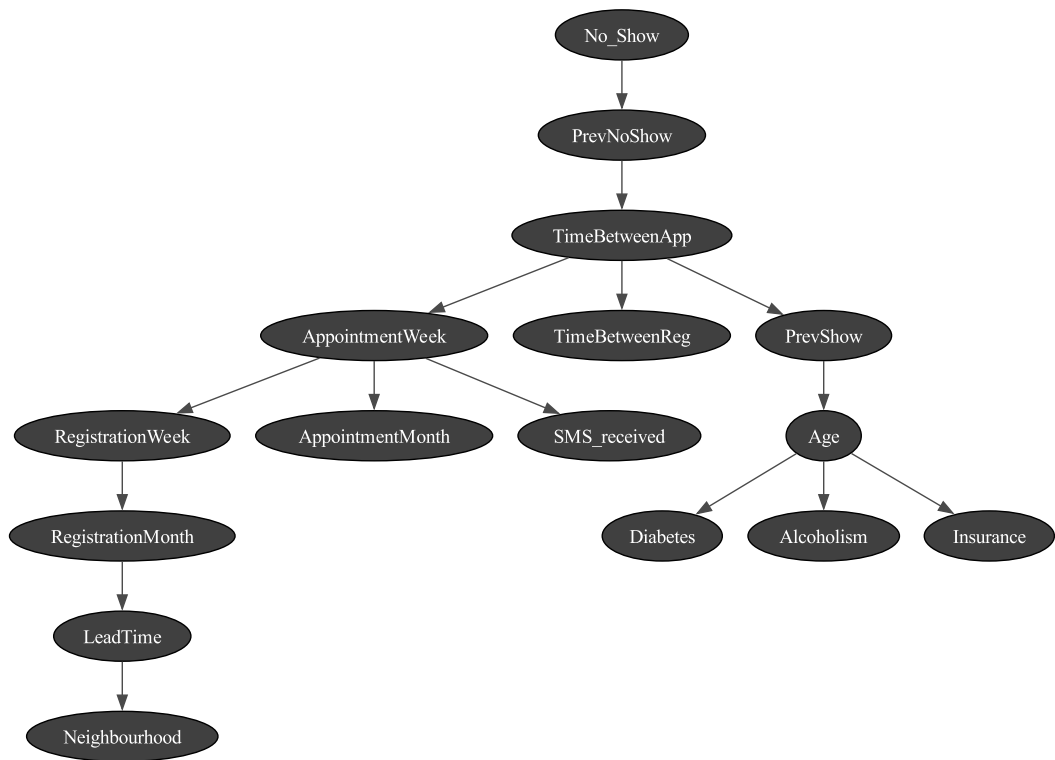


FIGURE 4. Bayesian belief network structure.

TABLE 5. Highest no-show neighborhoods.

Neighbourhoods	No Show	
	No	Yes
JARDIM CAMBURI	6252	1465
MARIA ORTIZ	4586	1219
ITARARA	2591	923
RESISTANCIA	3525	906
CENTRO	2631	703
JESUS DE NAZARETH	2157	696
JARDIM DA PENHA	3246	631
CARATOCRA	1974	591
TABUAZEIRO	2559	573
BONFIM	2223	550
ILHA DO PRCNCIPE	1734	532
ANDORINHAS	1741	521
SEO PEDRO	1933	515
SANTO ANDRA	2063	508
SANTA MARTHA	2635	496

resource allocation and enhancing patient attendance through targeted interventions.

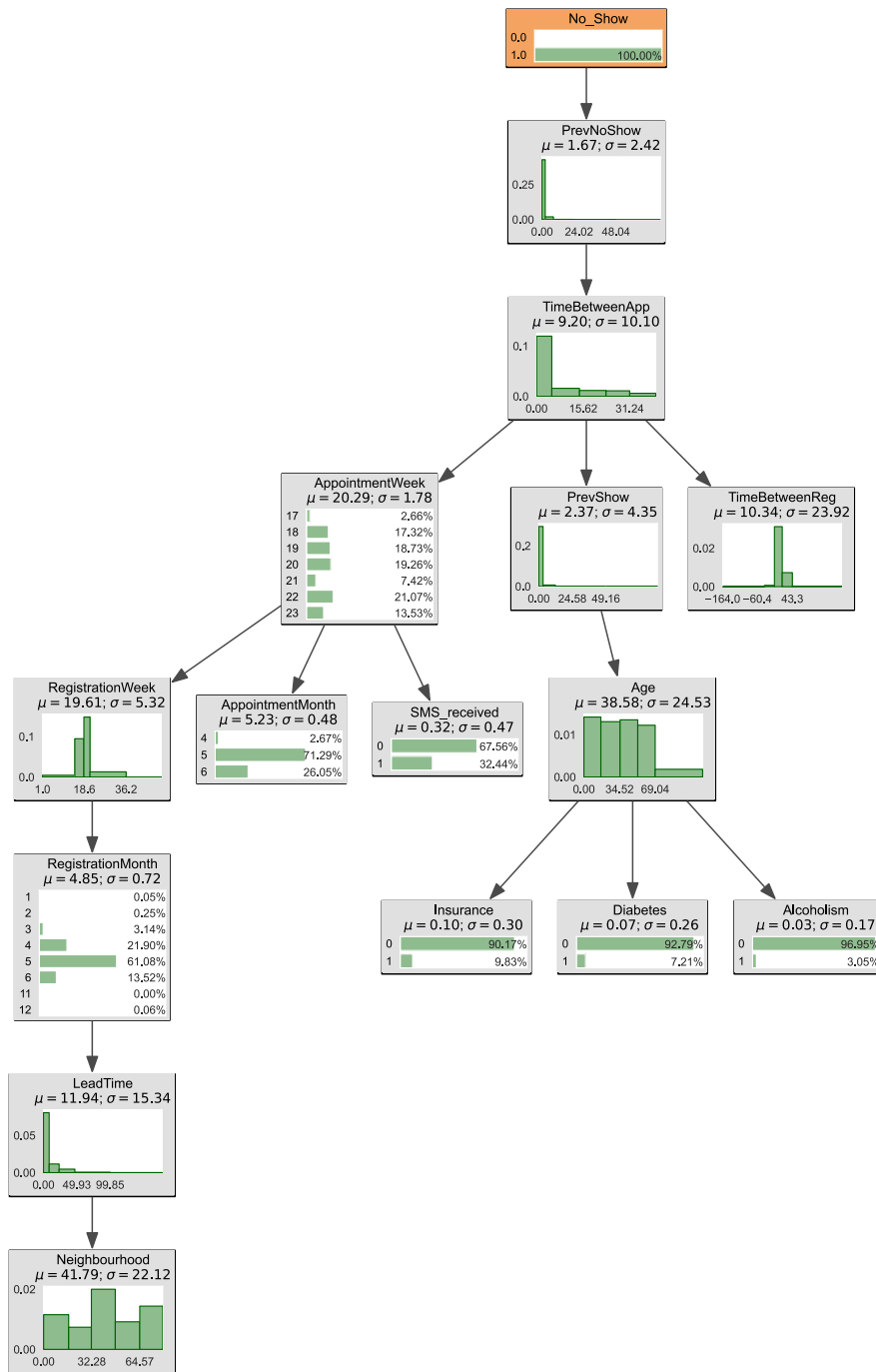
The investigated methodology for mitigating no-show risks has led to an enhanced appointment booking flow chart, as illustrated in Figure 7. This advanced procedure uses ML to identify no-show risk probabilities and implement targeted interventions based on the BBN inference diagram, focusing on the factors leading to no-shows.

To translate reductions in no-show probabilities into operational and economic terms, a simple cost-benefit calculation was applied. Evidence has indicated that each unused time slot has been estimated to represent approximately \$200 in lost physician revenue [4]. On this basis, a unit cost of \$200 per no-show was adopted. Clinical trials of reminder systems have reported a relative reduction in non-attendance of approximately 15% when patients are provided with SMS reminders compared with usual care. Under the assumption that the intervention under study would produce a comparable effect, this 15% reduction was multiplied by the total number of missed appointments in the dataset and subsequently by the \$200 cost per missed appointment. The resulting estimate indicated an annual saving of approximately \$670,000.

By leveraging patient data and behavioral patterns, the system predicts potential no-shows and offers dynamic rescheduling, personalized reminders, and intelligent over-booking. This data-driven approach optimizes resource utilization and enhances patient engagement, representing a shift from reactive to proactive healthcare management strategies. A detailed comparison between current practices and the suggested ones is presented in Table 6.

V. DISCUSSION

The study introduces a comprehensive knowledge-based framework designed to enhance the prediction of appointment no-shows, integrating refined ML algorithms with probabilistic graphical BBN models to provide an understanding



Inference in 41.87ms

FIGURE 5. Bayesian belief network inference.

of contributing factors and to facilitate targeted interventions. The methodology employed herein, which included feature selection, advanced data balancing techniques, and the construction of a BBN, has yielded demonstrably superior predictive performance when compared to existing approaches in the literature, as evidenced by the comparative analysis presented in Table 4. This notable improvement

in predictive accuracy, primarily marked by achieving an accuracy of 0.93 and an F1 score of 0.83, highlights the efficacy of the integrated framework in addressing the complex challenge of appointment non-attendance within healthcare settings.

One of the pivotal aspects of this research lies in the approach to feature selection, where multiple methods were

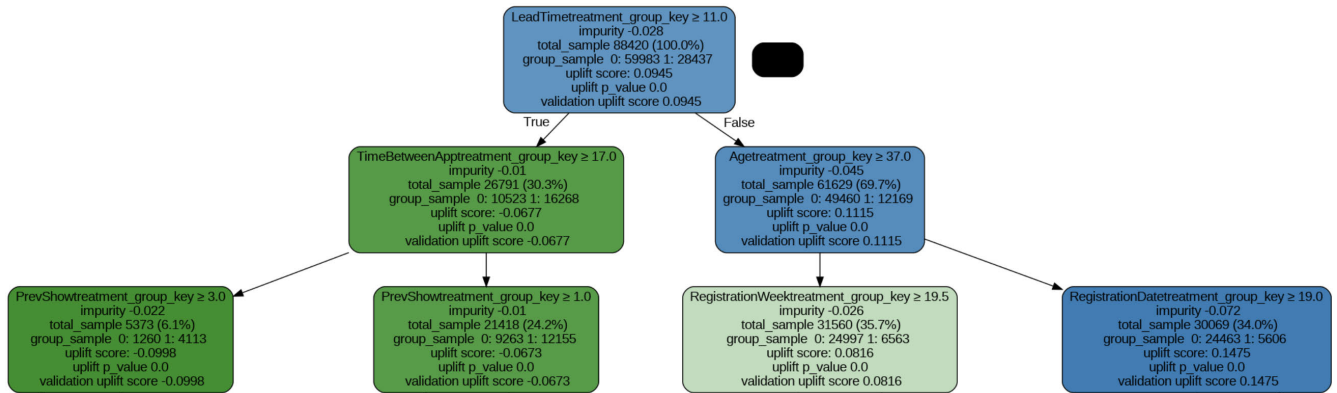


FIGURE 6. SMS reminder uplift 3-layer tree.

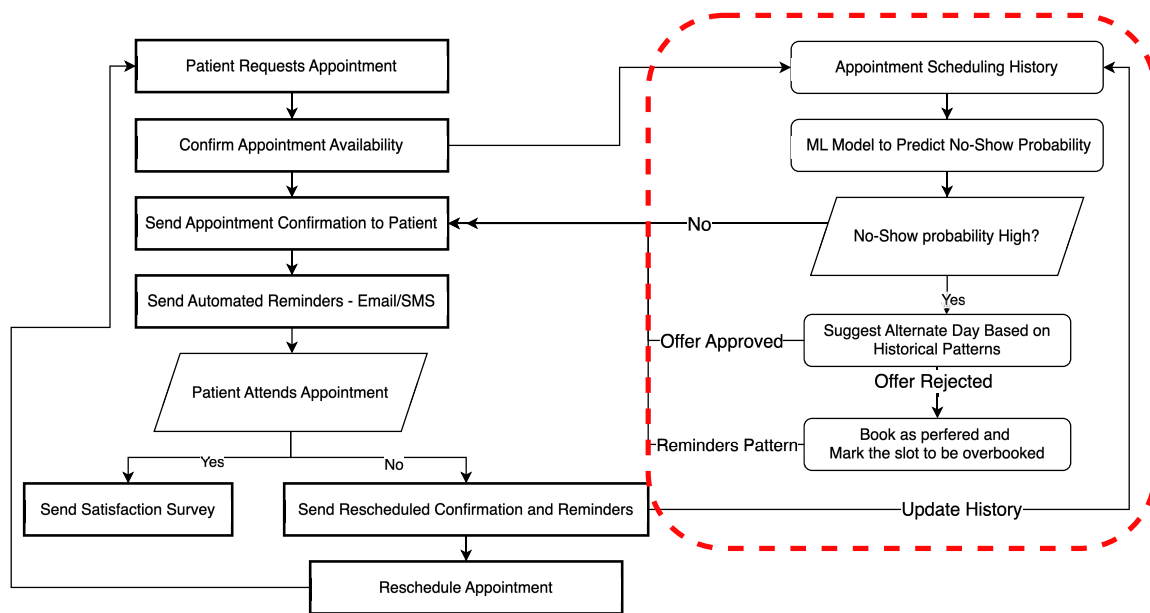


FIGURE 7. Current and proposed appointment booking practices.

systematically evaluated to identify the most significant predictors of no-show behavior. The selection of an optimal feature subset, subsequently utilized across both ML classifiers and the BBN, ensured consistency and interpretability across the diverse modeling approaches. The results, as detailed in Table 2, highlight the varying importance of different features across these selection methods, emphasizing the multifaceted nature of factors influencing appointment attendance. For instance, while ‘Age’ and ‘Neighbourhood’ were consistently identified as significant by multiple methods, the inclusion of temporal factors such as ‘Lead Time’ and ‘Time Between Appointments’ further enriched the predictive power of the models, reflecting the dynamic interplay of patient-related and appointment-related variables.

Furthermore, the class imbalance within the no-show dataset, characterized by a significantly higher proportion of attended appointments, necessitated the application of a

diverse array of data balancing techniques. The comprehensive evaluation of undersampling, oversampling, hybrid, and ensemble-based balancing methods, alongside cost-sensitive learning paradigms, was crucial in mitigating model bias and enhancing the predictive accuracy for the minority no-show class. The findings indicate that the combination of SFS with SMOTEEN for data balancing, coupled with an SR model, yielded the most robust performance, achieving an AUROC of 0.83 and a macro F1 score of 0.83. This particular combination not only demonstrates a strong predictive capability but also suggests that a balanced representation of both attended and missed appointments is critical for developing models that are both accurate and generalizable to real-world scenarios.

Beyond prediction, this study significantly advances the understanding of no-show phenomena through the construction and analysis of a BBN, causal inference, and

TABLE 6. Comparison of current and proposed appointment booking practices.

Factor	Current Approach	New Approach
Efficiency	Manual confirmations with automated reminders. The reactive system that doesn't anticipate no-shows. Relies heavily on manual intervention for rescheduling, which can lead to potential delays and inefficiencies due to a limited ability to optimize scheduling based on patient behavior or historical data.	Uses machine learning to predict no-shows and suggests alternate days or overbooking options. Proactively optimizes resource utilization and reduces the need for last-minute rescheduling.
Accuracy	Accurate for basic scheduling, but lacks predictive capabilities. May lead to more no-shows and underutilized time slots. Appointment confirmations are subject to human error during manual processing.	Leverages machine learning models to predict no-show probabilities, improving scheduling and resource allocation accuracy. Allows for more precise appointment management.
Consistency	Offers consistent basic functionality across all appointments with standard confirmation and reminder processes. However, manual interventions may lead to inconsistencies in handling rescheduling or cancellations.	Provides a higher level of consistency by applying predictive analytics to all bookings. Ensures uniform decision-making and resource allocation across the system.
Cost Effectiveness	Automates some processes, but results in higher costs due to unused time slots from no-shows and increased administrative work for rescheduling. Inefficient use of staff time for manual confirmations and rescheduling processes.	Reduces no-shows through predictive analytics and overbooking strategies, maximizing resource utilization. Leads to significant cost savings by reducing wasted time and improving operational efficiency.

treatment effects. The BBN optimized using the Chow-Liu structure learning method with k-means discretization provides a transparent and interpretable representation of the probabilistic relationships among various influencing factors. The network structure, visually represented in Figure 4, reveals intricate dependencies, such as the direct influence of 'PrevNoShow' on the 'No-Show' outcome, mediated by 'TimeBetweenApp'. This insight into causal pathways allows healthcare providers to move beyond simple correlations and to identify specific points of intervention. For example, the BBN analysis highlighted that factors such as distance from the hospital, insurance status, alcohol consumption history, and prevalence of chronic diseases are primary drivers of no-shows within specific neighborhoods, as detailed in Table 5. Such actionable intelligence is invaluable for designing geographically and demographically targeted strategies.

Moreover, the application of causal inference techniques, including ATE analysis and uplift modeling, provided quantitative evidence for the effectiveness of interventions, particularly SMS reminders. The ATE score of -0.149, indicating a 15% average decrease in no-show probability for individuals receiving SMS reminders, provides a compelling argument for their continued and optimized use. More importantly, the uplift tree model, illustrated in Figure 6, provides insight into heterogeneous treatment effects, identifying specific patient subgroups that benefit the most from

these reminders. For instance, patients with longer lead times or a history of multiple previous no-shows exhibited a more pronounced reduction in no-show probability upon receiving SMS reminders. This stratification of intervention effectiveness is crucial for optimizing resource allocation and maximizing the impact of patient engagement initiatives, moving towards a more personalized and proactive healthcare management paradigm.

Finally, the proposed methodology guides an enhanced appointment booking flowchart, as presented in Figure 7, which represents a paradigm shift from reactive to proactive healthcare management. This advanced procedure leverages ML to predict no-show risk probabilities and integrates targeted interventions based on the BBN inference diagram. The comparison between current and proposed practices, outlined in Table 6, highlights the potential for significant improvements in efficiency, accuracy, consistency, and cost-effectiveness. By dynamically adjusting scheduling, offering personalized reminders, and enabling intelligent overbooking based on predicted no-show probabilities, healthcare organizations can optimize resource utilization, reduce wasted appointments, and ultimately enhance patient access to care. This holistic approach, grounded in robust data analytics and probabilistic reasoning, offers a sustainable solution to a persistent challenge in healthcare delivery, paving the way for more efficient and patient-centric services.

VI. CONCLUSION

In conclusion, this study has successfully developed and validated a novel knowledge-based framework for predicting and mitigating appointment no-shows, a pervasive challenge within healthcare systems that significantly impacts resource allocation and patient care. By integrating advanced ML techniques with probabilistic modeling through a BBN, and treatment effect analysis through ATE and uplift modeling, this research has not only achieved superior predictive accuracy compared to existing methodologies but has also provided insights into the complex interplay of factors contributing to patient non-attendance. The application of feature selection methods, coupled with data balancing strategies, proved instrumental in overcoming data complexities and enhancing the generalizability and robustness of the predictive models.

Furthermore, the construction and analysis of the BBN offered a level of interpretability, explaining the complex causal relationships between diverse patient-related and appointment-related variables and the ultimate no-show outcome. This transparency is critical for healthcare administrators and practitioners, enabling them to move beyond mere statistical correlations and to pinpoint specific, actionable intervention points. The quantitative assessment of intervention effectiveness, particularly concerning SMS reminders through ATE analysis and uplift modeling, demonstrated the tangible benefits of targeted communication strategies and highlighted the importance of personalized approaches based on patient characteristics and historical behavior. These

findings highlight the potential for data-driven interventions to significantly reduce no-show rates and optimize healthcare delivery.

A. CONTRIBUTIONS

This study makes several significant contributions to both the academic literature and practical healthcare management. Firstly, it introduces a comprehensive predictive framework that integrates knowledge representation paradigms, offering a robust and accurate solution for identifying high-risk no-show patients. This framework's superior performance, as demonstrated by comparative benchmarks, signifies a notable advancement in the field of appointment no-show prediction. Secondly, the research provides a holistic understanding of the multifaceted factors influencing appointment attendance through the development and analysis of BBN, which not only captures complex probabilistic interactions but also offers a transparent and interpretable model for identifying causal pathways, a crucial aspect for effective intervention design. Thirdly, the study quantifies the effectiveness of specific interventions, such as SMS reminders, using causal inference techniques like ATE analysis and uplift modeling. This empirical evidence provides valuable insights for optimizing resource allocation and enhancing patient attendance rates through targeted strategies. Ultimately, the proposed enhanced appointment booking flowchart presents a practical and implementable solution that can transform current reactive healthcare management practices into proactive, data-driven systems, resulting in improved resource utilization, cost savings, and enhanced patient engagement and care quality.

B. IMPLICATIONS

The implications of this research are wide-ranging, impacting healthcare providers, policymakers, and researchers. For healthcare providers, the developed predictive framework offers a powerful tool for operational efficiency. By accurately identifying patients at high risk of missing appointments, resources can be proactively reallocated, reducing wasted time and maximizing clinic capacity. This leads to significant cost savings and improved patient flow. The interpretability of the BBN empowers healthcare professionals to understand the underlying reasons for no-shows, facilitating the design of more effective and personalized intervention strategies. For policymakers, the findings provide a strong evidence base for developing data-driven policies aimed at reducing healthcare inefficiencies and improving patient access to care. The demonstrated effectiveness of targeted interventions can inform public health initiatives and resource allocation decisions at a broader scale. Methodologically, this study exemplifies the power of integrating advanced analytical techniques to address complex real-world problems, suggesting that similar approaches could be applied to other domains beyond healthcare, such as optimizing resource management in education or transportation systems. This research thus serves as a foundational work, encouraging

further exploration and application of integrated analytical frameworks in diverse fields.

C. LIMITATIONS AND FUTURE WORK

Despite the contributions of this study, certain limitations should be acknowledged, and opportunities for future research remain open. The generalizability of the findings is limited, as the study primarily utilized an open-source dataset from Brazilian public healthcare providers. While comprehensive, the specific demographic and healthcare system characteristics of this dataset may limit the direct applicability of the findings to other geographical regions or healthcare contexts without further validation. Future work could involve testing the proposed framework on diverse datasets from various healthcare systems globally to assess its broader applicability and robustness. Additionally, while the study explored a wide array of feature selection and data balancing techniques, the dynamic nature of patient behavior and healthcare systems suggests that continuous monitoring and adaptation of these methods may be necessary. Future research could investigate the development of adaptive learning models that can continuously update their parameters and feature importance based on new incoming data, thereby maintaining optimal predictive performance over time.

Another area for future exploration involves integrating real-time data streams. The current framework, while robust, relies on historical data for prediction. Incorporating real-time patient information, such as current health status, transportation availability, or immediate communication responses, could further enhance the predictive accuracy and enable more dynamic and timely interventions. Furthermore, while the study demonstrated the effectiveness of SMS reminders, future research could delve deeper into the efficacy of other intervention modalities, such as personalized phone calls, community health worker outreach, or financial incentives, and explore optimal combinations of these strategies. Investigating the cost-effectiveness of different intervention mixes would also be a valuable addition. Finally, expanding the causal inference analysis to include a broader range of potential interventions and their heterogeneous effects across various patient subgroups would provide even more granular insights for precision healthcare management. This could involve developing more sophisticated uplift modeling techniques or exploring novel causal discovery algorithms to uncover previously unidentifiable intervention opportunities. Ultimately, the continuous refinement and expansion of this knowledge-based framework hold potential for revolutionizing appointment management and improving healthcare outcomes globally.

VII. DECLARATIONS

A. ETHICS APPROVALS

The analytical dataset was derived from a publicly accessible source that contains no direct identifiers; the provider had removed names, addresses, and contact information before

release. The study protocols adhered to all institutions' ethical standards and regulatory requirements.

B. DATA AVAILABILITY

This study utilizes publicly available, anonymized no-show data from Kaggle, collected from Brazilian public healthcare providers. The use of this dataset complies with Kaggle's terms of use and relevant ethical guidelines for secondary data analysis.

C. COMPETING INTERESTS

The authors declared no potential conflicts of interest concerning this article's authorship and/or publication.

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within healthcare settings. This approach has the potential to improve patient outcomes and enhance overall healthcare management across various applications.

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